Task 1:

Code:

Terminal

```
#!/bin/bash

greatest() {
        arr=("$@")
        max=${arr[0]}
        for i in "${arr[@]}"; do
                if (( i > max )); then
                        max=$i
                fi
        done
        echo $max
}

main() {
        read -p "Enter the number of elements: " n
        arr=()
        echo "Enter $n elements:"
        for ((i=0;i<n;i++)); do
                read num
                arr+=($num)
        done
        max=$(greatest "${arr[@]}")
        echo "Maximum element is : $max"
}

main
```

Output:

Terminal

```
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_05_01.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_05_01.sh
-rwxrwxr-x 1 bodike bodike 355 Aug 19 17:06 B24CS013_05_01.sh
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_01.sh -o B24CS013_05_01
Enter the number of elements: 5
Enter 5 elements:
1
9
7
5
8
Maximum element is : 9
bodike@bodike-chaithali:~/Scripts$
```

Task 2:

Code:

Terminal

```
#!/bin/bash

read -p "Enter filename: " file

if [[ -f "$file" ]]; then
        lines=$(wc -l < "$file")
        words=$(wc -w < "$file")
        chars=$(wc -m < "$file")
        echo "Number of lines: $lines"
        echo "Number of words: $words"
        echo "Number of characters: $chars"
else
        echo "404: file not found"
fi
```

Output:

```
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_05_02.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_05_02.sh
-rwxrwxr-x 1 bodike bodike 289 Aug 19 17:18 B24CS013_05_02.sh
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_02.sh -o B24CS013_05_02
Enter filename: B24CS013_05_01.sh
Number of lines: 26
Number of words: 56
Number of characters: 355
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_02.sh -o B24CS013_05_02
Enter filename: file.txt
404: file not found
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_05_02.sh
bodike@bodike-chaithali:~/Scripts$
```

Task 3:

Code:

Terminal

```
#!/bin/bash

read -p "Enter a URl: " url
regex="^https?:\/\/[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}.*$"

if [[ $url =~ $regex ]]; then
        status=$(curl -o /dev/null -s -w "%{http_code}" "$url")
        if [[ $status =~ ^2|3[0-9]{2}$  ]]; then
                echo "Valid and reachable link."
        else
                echo "Link format valid but unreachable."
        fi
else
        echo "Invaid link format."
fi
~
```

Output:

Terminal

```
bodike@bodike-chaithali:~/Scripts$ vi B24CS013_05_03.sh
bodike@bodike-chaithali:~/Scripts$ ls -l B24CS013_05_03.sh
-rwxrwxr-x 1 bodike bodike 348 Aug 19 18:37 B24CS013_05_03.sh
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_03.sh -o B24CS013_05_03
Enter a URl: https://www.google.com
Valid and reachable link.
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_03.sh -o B24CS013_05_03
Enter a URl: https://ac.in
Link format valid but unreachable.
bodike@bodike-chaithali:~/Scripts$ ./B24CS013_05_03.sh -o B24CS013_05_03
Enter a URl: www.google.com
Invaid link format.
bodike@bodike-chaithali:~/Scripts$ ls
B24CS013_05_01.sh  B24CS013_05_02.sh  B24CS013_05_03.sh
bodike@bodike-chaithali:~/Scripts$
```